

Volcanic Seismic Event Classification based on CNN Architectures

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Abstract—This paper explores the use of convolutional neural networks architectures in the context of volcano seismic events classification through the use of grayscale spectrogram images of long-period and volcano-tectonic seismic events. We combined three different architectures with a set of hyper-parameter configurations that produced 720 classification models, which were able to learn the morphological pattern described by the grayscale spectrogram images of seismic events. Downscaling of all grayscale spectrogram images was used to reduce the computation time for each model without losing performance and avoiding any overfitting. The explored architectures provided good results in terms of the area under the receiver operating characteristic curve scores. However, when considering the accuracy scores in the selection process, the best model to classify grayscale spectrogram images of both types of seismic events was the CNN3 architecture with a (3×3) convolutional and pool kernel size, same padding type and 1×10^{-4} as learning rate, which achieved an area under the receiver operating characteristic curve and accuracy values of 0.94 and 94.13%, respectively.

Keywords—volcanic seismic event classification, machine learning classifier, CNN based classifiers, spectrogram images.

I. INTRODUCTION

Monitoring the volcanic activity is an indispensable factor in reducing the associated effects that could be produced by volcanic eruptions, specially in highly populated cities located near active volcanoes such as Mexico City (Mexico) close to the Popocatepetl volcano or Tokyo (Japan) near to Mt. Fuji or Naples (Italy) close to Vesuvius or Seattle (USA) close to Mount Rainier among others [1]. In fact this type of natural phenomena has been responsible directly and indirectly for thousands of deaths since the year 1500 [2]. Usually, volcanoes are monitored by analysing their seismic signatures [1], [3].

A wide range of approaches using machine learning have been developed in recent years to address the problem of volcano seismic events classification, such as: random forest (RF) [4], [5], decision trees (DT) [6], hidden Markov models

(HMM) [7], Gaussian mixture models (GMM) [8], support vector machine (SVM) methods [5], [9]–[11], boosting strategies [12] and artificial neural networks (ANN), specially multi-layer perceptrons models [5], [13]–[15]. However, the majority of these methods require feature calculation and selection stages to minimize the volume of information used to feed the models, which helps them to avoid overfitting during model training. This extra workload makes some models non-practical to be used for the classification of spectrogram images of seismic events. The convolutional neural networks (CNN) are special ANN architectures that are gaining more attention in image analysis contexts [16], [17]. They avoid using intermediate fully connected layers to employ pooling ones and thus to optimize the information pass-through from layer to layer. Lately, there is evidence of deep CNN models used to classify spectrogram images of seismic events with successful performance [18]. However, the classification of volcano seismic events remains a challenging and useful problem to solve.

This work proposes a CNN classifier based exploration to tackle the problem of seismic events classification, focusing on long-period (LP) and volcano-tectonic (VT) seismic events recorded from the Cotopaxi Volcano in Ecuador. This work is based on a combination of three different CNN models that learn the morphological pattern described in the grayscale spectrogram images of both types of events. Then, the most accurate model per architecture is selected, and a final decision rule determines which model should be considered as the best solution. We selected CNN based classifiers instead of other machine learning classifiers (MLCs) mainly because they do not involve any feature calculation/selection stages and have demonstrated a high capability to handle image analysis problems [16], [17].

The remainder of this paper is organized as follows: the materials and methods section presents the experimental spectrogram images dataset used for our experimentation, as well as, the considered CNN architecture based classifiers and the

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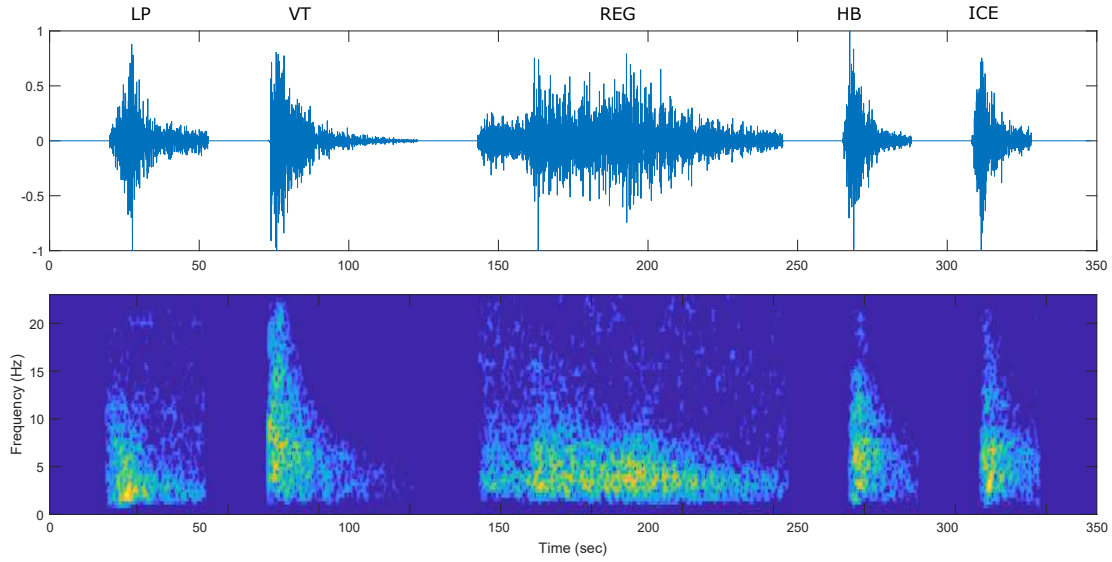


Fig. 1. Wrapped examples from the *MicSigVI* dataset: time-domain signals (top) and their respective spectrogram (bottom). The time signals were normalized by their maximum absolute value (Figure taken from [19]).

experimental setup design used in this work. The results and discussion section presents an exploratory comparison based on the obtained area under the receiver operating characteristic curve (AUC) scores for each CNN based models and the accuracy (ACC) scores against the state of art-based methods. Finally, conclusions and future work are drawn in the last section.

II. MATERIALS AND METHODS

A. Experimental dataset

We used the (*MicSigVI*) dataset from the ESeismic repository¹, which is the first annotated Ecuadorian volcano seismic public repository containing several samples recorded at the Cotopaxi volcano [19]. The *MicSigVI* is composed by a total of 1187 seismic records from two different seismic stations (VC1 and BREF) installed at the Cotopaxi volcano. The records were acquired by a three-axial seismometer (CMG-40T Guralp) with a sensitivity of $1600 \text{ V/m}\cdot\text{s}^{-1}$, using a 24-bit analog to digital converter (Geotech Smart 24D digitizer) sampling at 50 Hz. Each record in the dataset contains one independent seismic event, which was manually identified and labeled by analysts at the IGEPN considering its waveform, spectrogram, and location.

The dataset contains samples distributed in five classes corresponding to: LP, VT, regional (REG), hybrid (HB), and icequakes (ICE). Some examples of seismic events inside this dataset are shown in Fig. 1. Due to the small number of samples from REG (27 instances), HB (8 instances), and ICE (7 instances) events available within the *MicSigVI* dataset, we only considered the use of LP and VT events classes for this study. The selected samples belong to the same seismic

station (VC1) to guarantee the same acquisition protocol and to avoid mixed signals. Therefore, the formed experimental dataset contains a total of 668 seismic records distributed in 587 cases of LP and 81 cases of VT seismic events classes. We kept the imbalanced representation of the classes to exploit the MLCs performance in a real-life scenario.

B. CNN Classifier

CNN classifiers have been widely adopted as a reliable MLC for developing computer vision and image recognition systems [20], [21]. The traditional structure consists of a sequential workflow starting at the convolutional layer, which generates the set of features extracted from the inputs by convolving the original image (input) with some filters. The features are represented as matrices that are then passed to an activation layer, usually with the sigmoid function. Afterward, a pooling layer is used for the purpose of sampling and performing feature reduction through the use of max pooling, min pooling, or average pooling methods. In the last step, a fully connected or dense layer and an output activation layer with a softmax function (equivalent to the sigmoid function for binary cases) are included to provide the classification capabilities of the model [20].

We adopted the general structure of the CNN architecture to build three different classifiers, as it is shown in Fig. 2. The classifier design was mainly focused on the variation of the number of filters. Therefore, CNN1, CNN2, and CNN3 classifiers were formed with a total of 8, 16, and 32 filters, respectively. Exploring the classification space under these variations allow us to better understand the relationship between the model performance and its complexity.

C. Experimental setup

The designed setup involved the spectrogram images pre-processing, training and test partitions, CNNs configuration,

¹ESeismic repository was provided by courtesy of the Instituto Geofísico de Escuela Politécnica Nacional (IGEPN) and collaborators, and it is available at http://www.igepn.edu.ec/eseismic_web_site/index.php, where you should register and fill a disclaimer agreement in order to obtain the data.

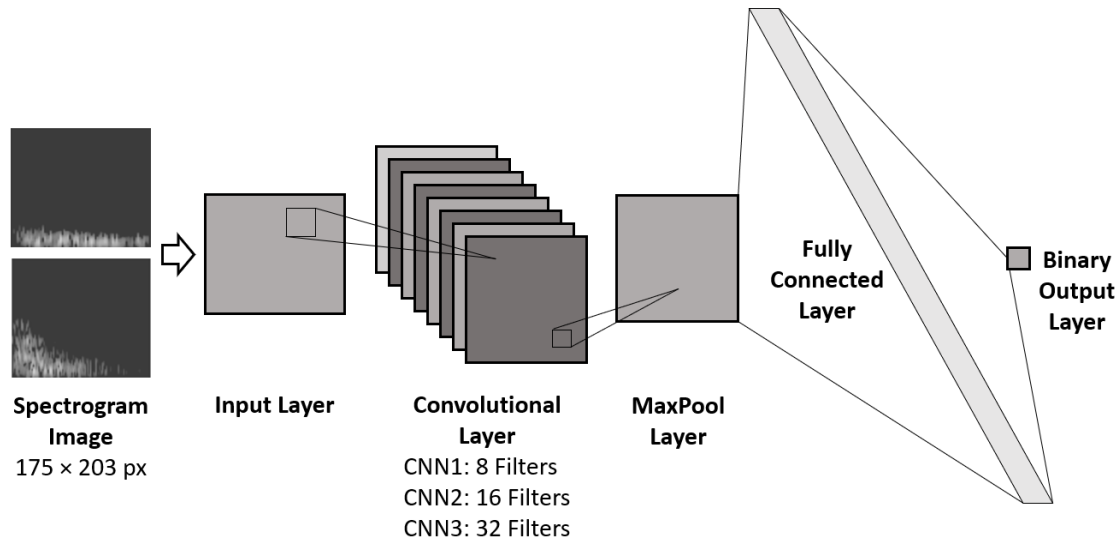


Fig. 2. CNN structure used in this work, from left to right: input layer, convolutional layer with its filters (one set of filters per model), and activation layer with sigmoid activation functions, max-pooling layer, fully connected layer (flattened) and a binary activation layer (output) with a sigmoid function.

assessment metrics, and selection criteria, which are essential aspects to be described next.

1) *Spectrogram images preprocessing*: Spectrogram images from each seismic records in the experimental dataset were generated using the short-time Fourier transform (STFT) [22] over the original signal. This process used a Kaiser window [23] with 1.5 s sampling window size and 75% of overlapping to avoid losing information in the boundary of the window and thus, maximizing the energy in the main part of the signal. Since the spectrogram images are in RGB (red, green, blue) space and the color information is irrelevant in the context of volcano events pattern analysis, we first transformed them into grayscale space to reduce the information channel of each pixel in the image and thus, to reduce the input information to the classifiers.

Also, the spectrogram image dimensions were downsampled to a 25% (175×203) of the original image sizes (701×813) to reduce the number of pixels per image and to further decrease the computational requirements of the learning models. The experimental dataset provides spectrogram images free of noise, thus, the downscaling operation does not affect the seismic event pattern presented on each image. The downsampled operation has been well developed and employed in other research works [24], [25] with successful results. Moreover, the pixel values of each spectrogram image were normalized using the min-max method [26] for bringing them into the range between -0.5 to 0.5 in order to avoid data dispersion.

2) *Training and test partitions*: We applied five-times the stratified 10-fold cross-validation method [27] before the classification stage to form disjoint training and test partitions, and to ensure the observation ratio between both types of events along all folds. In this way, individual CNN models will be trained on different training sets and, thus, learning from different input space representations. Testing on these

different sets leads to variability in the resulting classification for individual samples.

3) *CNNs configuration*: For all of the three developed models, the CNN1 with 8 filters, CNN2 with 16 filters and CNN3 with 32 filters, the convolutional kernel was optimized using a (3×3) , (7×7) and (15×15) dimensions. The padding types were tuned to *no_padding* and *same*, which means whether or not to add padding such that the resulting filter retains the dimensions of the input image. The learning rate was optimized in the range from 1×10^{-4} to 5×10^{-4} with increment steps of 2×10^{-4} . The max-pooling kernel size used a (3×3) and (2×2) dimensions with a stride of 3 and 2, respectively. The number of iterations (epochs) varied from 50 to 500 with increment steps of 50 units. Finally, the models used the *adam* optimizer, which is based on the adaptive estimation of lower-order moments and was designed to combine the advantages of the well-known optimizers AdaGrad and RMSProp [28].

4) *Assessment metrics*: The classification performance of all employed CNN models was primarily based on the AUC of the receiving operating characteristic curve. The ACC metric was computed for both the tiebreaker in AUC scores and to support state of the art based comparisons. Besides, all the statistical comparison in terms of AUC and ACC metrics was conducted using the Wilcoxon statistical test, which is a non-parametric alternative test to the paired *t*-test. This test ranks the differences in performances of two MLCs [29], providing a fair comparison among them, and therefore a reasonable selection of the best classification model. We used a significance decision level value of 5% ($\alpha = 0.05$) for a two-tailed test [30] on all comparisons.

5) *Selection criteria*: Since the CNN1, CNN2, and CNN3 classifiers explored several model configurations, it was mandatory to select the best model according to a decision rule. Despite the lack of a universal rule to select the best

MLC, we stated a “rule of gold” for the selection based on the following criteria: (1) the highest AUC score statistically per classifier and, (2) if there was a tie rating performance in AUC scores, the one that reaches the highest ACC value statistically ($p < 0.05$) was preferred. It should be noted that selection of more than one model is possible depending on whether or not there is an AUC based significant difference among the classification models. Thus, the ACC scores will determine the final selection.

The implementation of all classifiers was done in Python language version 3.6.9 [31] with *scikit-learn* (*SKlearn*) library [32] and Keras [33] (MXNET backend [34]) CNN classifiers.

III. RESULTS AND DISCUSSION

A total of 720 CNN based models were evaluated on the experimental dataset containing 668 spectrogram images. The straightforward statistical comparison based on the mean of AUC performance over 50 runs highlighted interesting results for the classification of LP and VT seismic events, as described next:

A. Performance evaluation

Agreeable to the first selection criterion, a total of 3 out of 720 classification models were obtained after exploring the whole space of classifiers. According to the Wilcoxon statistical test at $p < 0.05$, the best model using the CNN1 classifier was formed by the (3×3) convolutional and pool kernel size, *no_padding* type, 50 epochs and 3×10^{-4} as learning rate, reaching AUC and ACC values of 0.95 and 91.89%, respectively. The best model provided by the CNN2 classifier was composed by a (3×3) convolutional kernel size, (2×2) pool kernel size, *same* padding type, 100 epochs and 1×10^{-4} as learning rate, reaching AUC and ACC values of 0.94 and 93.20%, respectively. For the CNN3 classifier, the best model was formed by a (3×3) convolutional and pool kernel size, *same* padding type, 150 epochs and 1×10^{-4} as learning rate, reaching AUC and ACC values of 0.94 and 94.13%, respectively.

The three selected models reached similar statistically AUC scores ($p < 0.05$) performance without incurring any overfitting during the training step, as shown in Fig. 3. Thus, it is possible to state that any of them could be used to handle the problem of LP and VT seismic events classification based on their grayscale spectrogram images. However, in terms of ACC values, the CNN3 classifier based models provided the best results statistically ($p < 0.05$). Intuitively, this situation can be explained for the extra features being learnt from the patterns in the grayscale spectrogram images by the network through its 32 filters. As drawback, the CNN3 classifier based models demand more interactions (epochs) to converge to the minimum error during the training and validation process (see Fig. 3, bottom row). Despite the good performance of the three considered models and according to the second selection criterion, the best classifier for the problem at hand was the model based on the CNN3 architecture.

TABLE I
COMPARISON BASED ON THE ACC BETWEEN RELATED PREVIOUS WORKS
AVAILABLE IN THE LITERATURE AND THE BEST SELECTED MODEL
PRODUCED IN THIS WORK.

Method	Number of samples	Computed features	Spectrogram images	ACC* (%)
ANN [6]	914	6	no	97
DT [6]	914	3	no	96
ANN [5]	637	17	no	95
RF [5]	637	17	no	93
Linear SVM [11]	914	5	no	97
ANN [15]	1033	8	no	94
HMM [7]	-	5	no	90
GMM [8]	667	2	no	94
CNN3 model	668	-	yes	94

ACC - accuracy; *values rounded to the closest integer

B. State-of-the-art comparison

Regarding the classification performance, it was not possible to make a statistically direct comparison against previously developed methods in the literature, since all of them included a feature calculation and selection stage instead of working directly on the spectrogram images like in our work. However, Table I shows a non statistical comparison based only on reported ACC scores by the state of the art methods. Since, these methods are not statistically comparable we include important works that not necessary use the same database (please refer to ANN [15] and HMM [7] works for more details).

From Table I, it is possible to notice that the selected CNN3 based model was able to reach a competitive ACC score when compared to previously developed methods. Except for the RF and feed-forward back-propagation based ANN developed in [5], which also worked on a dataset containing signal overlapped. The remaining related works performed the classification on a controlled dataset environment (balanced dataset, signals without overlapping, etc.). Also, it should be noted that all the state-of-art methods computed and reduced the feature space in some way to decrease the heavy load of information before feeding the MLCs, but, semantically, they increased the processing time of the process. Although the CNN3 model is computationally more expensive than some of the state of the art MLCs. It represents a simpler, straightforward workflow, and an execution time-saver since it avoids any feature selection/calculation stages.

IV. CONCLUSIONS AND FUTURE WORK

This work explored CNN classifiers to tackle the problem of seismic events classification, especially on grayscale spectrogram images related to LP and VT events from the Cotopaxi Volcano. We used a combination of three different CNN architectures with a set of hyper-parameter configurations that produced 720 classification models, which were able to learn the morphological pattern described by the grayscale spectrogram images of both types of events. Downscaling of all grayscale spectrogram images was used to reduce the computation time required by each model without losing AUC

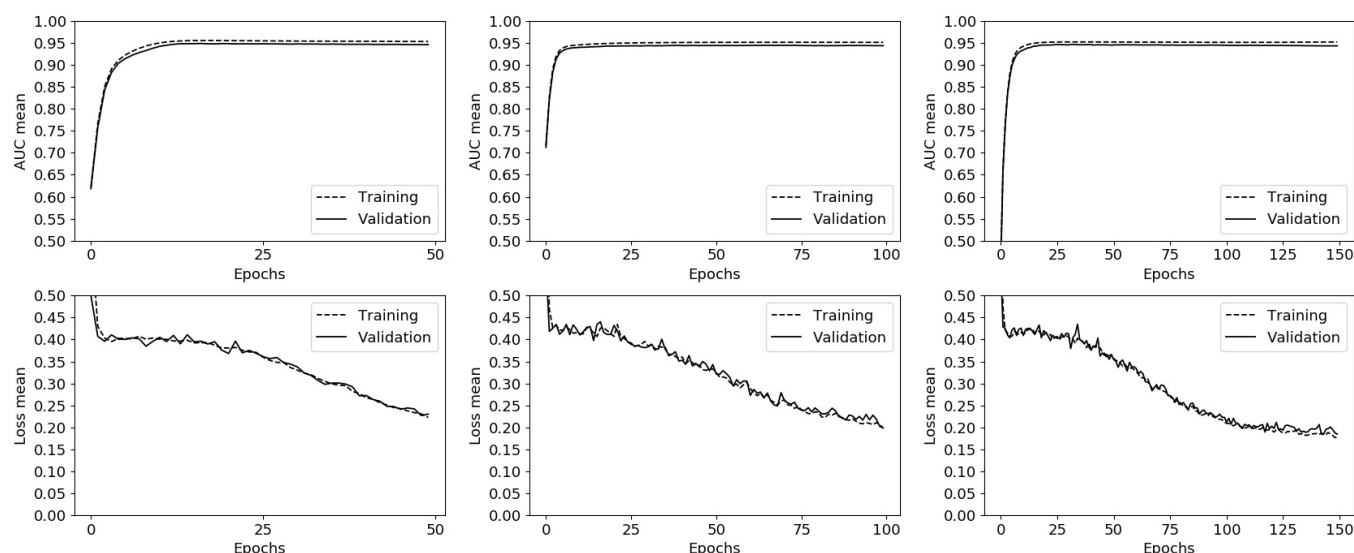


Fig. 3. Performance results of the best-selected model from CNN1 (left), CNN2 (center) and CNN3 (right) classifiers based on the AUC (top row) and loss function (bottom row) metrics in the training and validation sets (top row).

and ACC performances, avoiding any overfitting. The three explored CNN architectures provided good results in terms of AUC scores. However, the CNN3 based models were much better when considering the ACC scores in the selection process. Therefore, the best model to classify grayscale spectrogram images of LP and VT seismic events was the CNN3 model with a (3×3) convolutional and pool kernel size, *same* padding type and 1×10^{-4} as learning rate, which reached an AUC and ACC values of 0.94 and 94.13%, respectively.

As future work, we plan (1) to incorporate new classification models by including other MLCs such as radial basis function networks, (2) to explore other preprocessing techniques of the original *MigSigVI* dataset, as well as, (3) to expand the hyperparameters configurations to understand the performance limits of the developed models.

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REFERENCES

- [1] H.-U. Schmincke, "Volcanic hazards, volcanic catastrophes, and disaster mitigation," in *Volcanism*. Springer, 2004, pp. 229–258.
- [2] R. I. Tilling, "Hazards and climatic impact of subduction-zone volcanism: A Global And Historical Perspective," *Washington DC American Geophysical Union Geophysical Monograph Series*, vol. 96, pp. 331–335, 1996.
- [3] M. Malfante, M. Dalla Mura, J.-P. Métaixian, J. I. Mars, O. Macedo, and A. Inza, "Machine learning for volcano-seismic signals: Challenges and perspectives," *IEEE Signal Processing Magazine*, vol. 35, no. 2, pp. 20–30, 2018.
- [4] M. Rodgers, P. Smith, D. Pyle, and T. Mather, "Waveform classification and statistical analysis of seismic precursors to the July 2008 vulcanian eruption of soufrière hills volcano, montserrat," in *EGU General Assembly Conference Abstracts*, vol. 18, 2016.
- [5] N. Pérez, P. Venegas, D. Benítez, R. Lara-Cueva, and M. Ruiz, "A new volcanic seismic signal descriptor and its application to a dataset from the cotopaxi volcano," *IEEE Transactions on Geoscience and Remote Sensing*, 2019, . Manuscript accepted for publication.
- [6] R. Lara-Cueva, E. V. Carrera, J. F. Morejon, and D. Benítez, "Comparative analysis of automated classifiers applied to volcano event identification," in *2016 IEEE Colombian Conference on Communications and Computing (COLCOM)*, April 2016, pp. 1–6.
- [7] M. C. Benítez, J. Ramirez, J. C. Segura, J. M. Ibanez, J. Almendros, A. Garcia-Yeguas, and G. Cortes, "Continuous hmm-based seismic-event classification at deception island, antarctica," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 45, no. 1, pp. 138–146, Jan 2007.
- [8] P. Venegas, N. Pérez, D. Benítez, R. Lara-Cueva, and M. Ruiz, "Combining filter-based feature selection methods and gaussian mixture model for the classification of seismic events from cotopaxi volcano," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 12, no. 6, pp. 1991–2003, 2019.
- [9] M. Curilem, J. Vergara, C. San Martin, G. Fuentealba, C. Cardona, F. Huenupan, M. Chacón, M. S. Khan, W. Hussein, and N. B. Yoma, "Pattern recognition applied to seismic signals of the llaima volcano (chile): An analysis of the events' features," *Journal of Volcanology and Geothermal Research*, vol. 282, pp. 134–147, 2014.
- [10] B. Apolloni *et al.*, "Support vector machines and mlp for automatic classification of seismic signals at stromboli volcano," in *Neural Nets WIRN09: Proceedings of the 19th Italian Workshop on Neural Nets, Vietri Sul Mare, Salerno, Italy May 28-30 2009*, vol. 204. IOS Press, 2009, p. 116.
- [11] R. A. Lara-Cueva, D. S. Benítez, E. V. Carrera, M. Ruiz, and J. L. Rojo-Álvarez, "Automatic recognition of long period events from volcano tectonic earthquakes at cotopaxi volcano," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 54, no. 9, pp. 5247–5257, Sept 2016.
- [12] P. Venegas, N. Pérez, D. S. Benítez, R. Lara-Cueva, and M. Ruiz, "Building machine learning models for long-period and volcano-tectonic event classification," in *2019 IEEE CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON)*. IEEE, 2019, pp. 1–6.
- [13] H. Langer, S. Falsaperla, T. Powell, and G. Thompson, "Automatic classification and a-posteriori analysis of seismic event identification at soufriere hills volcano, montserrat," *Journal of volcanology and geothermal research*, vol. 153, no. 1-2, pp. 1–10, 2006.

- [14] S. Scarpetta, F. Giudicepietro, E. C. Ezin, S. Petrosino, E. Del Pezzo, M. Martini, and M. Marinaro, "Automatic classification of seismic signals at mt. vesuvius volcano, italy, using neural networks," *Bulletin of the Seismological Society of America*, vol. 95, no. 1, pp. 185–196, 2005.
- [15] G. Curilem, J. Vergara, G. Fuentealba, G. Acuña, and M. Chacón, "Classification of seismic signals at villarrica volcano (chile) using neural networks and genetic algorithms," *Journal of Volcanology and Geothermal Research*, vol. 180, no. 1, pp. 1 – 8, 2009. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0377027308006355>
- [16] H.-C. Shin, H. R. Roth, M. Gao, L. Lu, Z. Xu, I. Nogues, J. Yao, D. Mollura, and R. M. Summers, "Deep convolutional neural networks for computer-aided detection: Cnn architectures, dataset characteristics and transfer learning," *IEEE transactions on medical imaging*, vol. 35, no. 5, pp. 1285–1298, 2016.
- [17] R. Chauhan, K. K. Ghanshala, and R. Joshi, "Convolutional neural network (cnn) for image detection and recognition," in *2018 First International Conference on Secure Cyber Computing and Communication (ICSCCC)*. IEEE, 2018, pp. 278–282.
- [18] M. Curilem, J. P. Canário, L. Franco, and R. A. Rios, "Using cnn to classify spectrograms of seismic events from llaima volcano (chile)," in *2018 International Joint Conference on Neural Networks (IJCNN)*, July 2018, pp. 1–8.
- [19] N. Pérez, D. Benítez, F. Grijalva, R. Lara-Cueva, and M. Ruiz, "ESeismic: Towards an Ecuadorian Volcano Seismic Repository," *Journal of Volcanology and Geothermal Research*, 2020, Submitted for publication.
- [20] S. Albawi, T. A. Mohammed, and S. Al-Zawi, "Understanding of a convolutional neural network," in *2017 International Conference on Engineering and Technology (ICET)*. IEEE, 2017, pp. 1–6.
- [21] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *European conference on computer vision*. Springer, 2014, pp. 818–833.
- [22] A. V. Oppenheim, "Applications of digital signal processing," *Englewood Cliffs, NJ, Prentice-Hall, Inc.*, 1978. 510 p, 1978.
- [23] J. F. Kaiser, "Nonrecursive digital filter design using the i_0 -sinh window function," in *Proc. 1974 IEEE International Symposium on Circuits & Systems, San Francisco CA, April, 1974*, pp. 20–23.
- [24] E. Kesim, Z. Dokur, and T. Olmez, "X-ray chest image classification by a small-sized convolutional neural network," in *2019 Scientific Meeting on Electrical-Electronics & Biomedical Engineering and Computer Science (EBBT)*. IEEE, 2019, pp. 1–5.
- [25] W. Quan, K. Wang, D.-M. Yan, and X. Zhang, "Distinguishing between natural and computer-generated images using convolutional neural networks," *IEEE Transactions on Information Forensics and Security*, vol. 13, no. 11, pp. 2772–2787, 2018.
- [26] Y. K. Jain and S. K. Bhandare, "Min max normalization based data perturbation method for privacy protection," *International Journal of Computer & Communication Technology*, vol. 2, no. 8, pp. 45–50, 2011.
- [27] F. G. López, M. G. Torres, B. M. Batista, J. A. M. Pérez, and J. M. Moreno-Vega, "Solving feature subset selection problem by a parallel scatter search," *European Journal of Operational Research*, vol. 169, no. 2, pp. 477–489, 2006.
- [28] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- [29] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine learning research*, vol. 7, no. Jan, pp. 1–30, 2006.
- [30] M. Hollander, D. A. Wolfe, and E. Chicken, *Nonparametric statistical methods*. John Wiley & Sons, 2013, vol. 751.
- [31] Python Core Team, *Python 3.6.9: A dynamic, open source programming language.*, Python Software Foundation, 2019. [Online]. Available: <https://www.python.org/>.
- [32] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [33] F. Chollet *et al.*, "Keras," <https://keras.io>, 2015.
- [34] T. Chen, M. Li, Y. Li, M. Lin, N. Wang, M. Wang, T. Xiao, B. Xu, C. Zhang, and Z. Zhang, "Mxnet: A flexible and efficient machine learning library for heterogeneous distributed systems," *arXiv preprint arXiv:1512.01274*, 2015.